Title: SOC lab report

Internship: SoC_task01_Report
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Date: 02-01-2026
Company: CyArt Tech

Task01. Objective

The objective of this task is to:

- Analyze Windows Security Event Logs to detect failed login attempts.
- Identify brute-force attack indicators.
- Export and analyze logs for investigation.
- Perform browser history forensic analysis using Eric Zimmerman tools.
- Identify suspicious or test URLs from Chrome browsing data.

1.2. Tools Used

Built-in Tools

- Windows Event Viewer
- wevtutil (Windows CLI)

1.3. System Setup

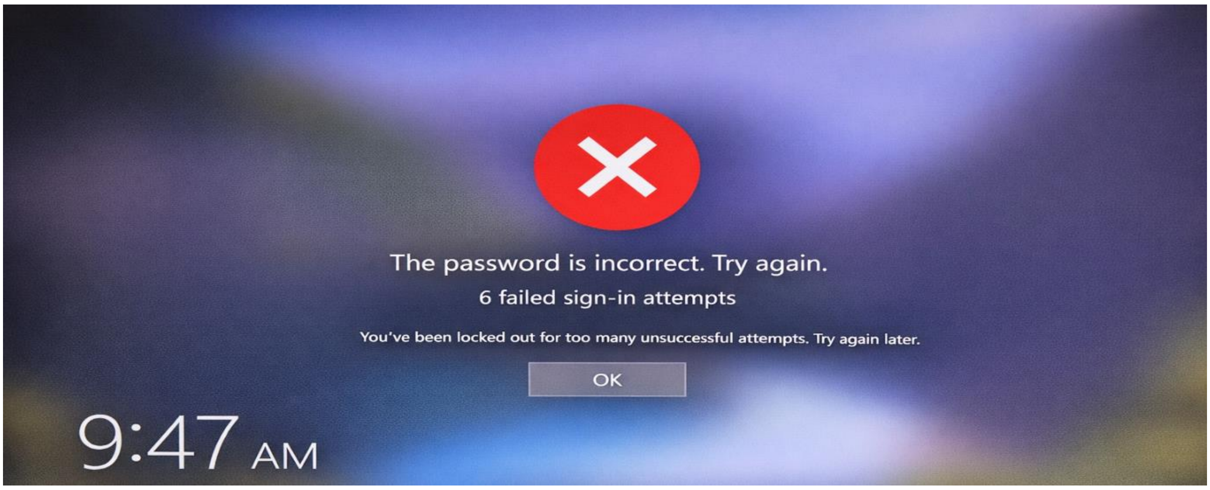
- Operating System: Windows 10 (Virtual Machine)
- User Account: Local Administrator
- Browser: Google Chrome
- Log Source: Windows Security Logs, Chrome History Database

1.4. Task – Failed Login Analysis (Event ID 4625)

- **Generating Failed Login Events**

Multiple failed login attempts were generated by intentionally entering incorrect passwords on the Windows login screen. These attempts triggered **Event ID 4625**, which records failed logon activities.

Screenshot 1: Failed login attempts on lock screen

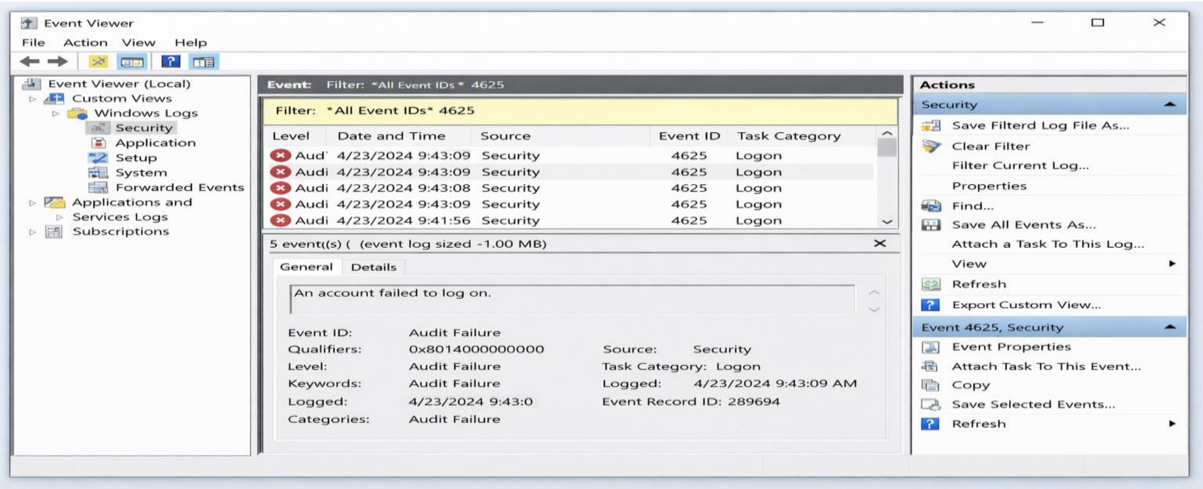


- **Event Viewer Log Analysis**

Steps followed:

1. Opened Event Viewer (eventvwr.msc)
2. Navigated to:
   Windows Logs → Security
3. Applied filter for **Event ID 4625**

Screenshot 2: Event Viewer filtered for Event ID 4625

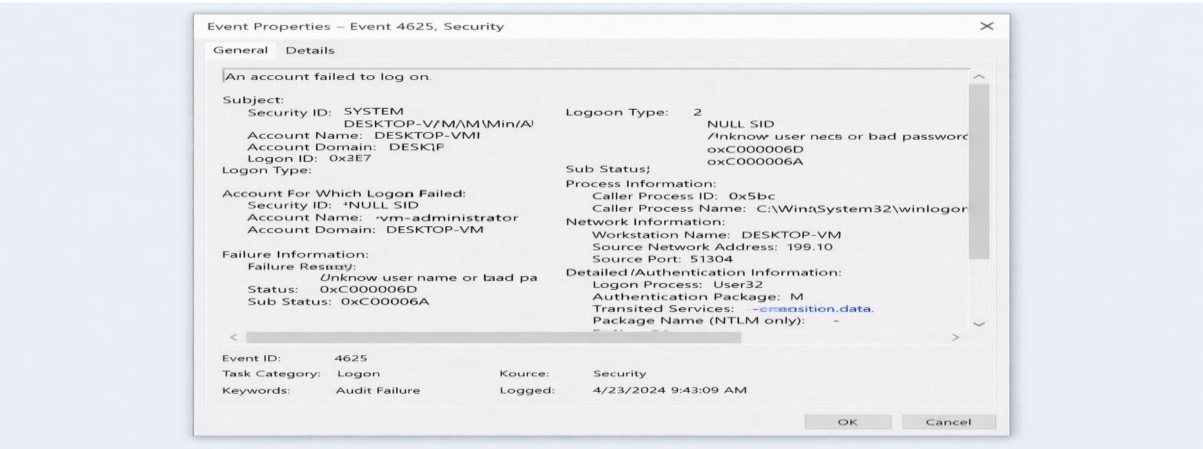


- **Log Details Observed**

Key fields analyzed:

- Account Name
- Logon Type
- Failure Reason
- Source Network Address
- Time of Attempt

Screenshot 3: Detailed view of Event ID 4625



1.5 Brute-Force Indicators

The following indicators suggest a brute-force attempt:

- Multiple failed login attempts
- Same user account targeted repeatedly

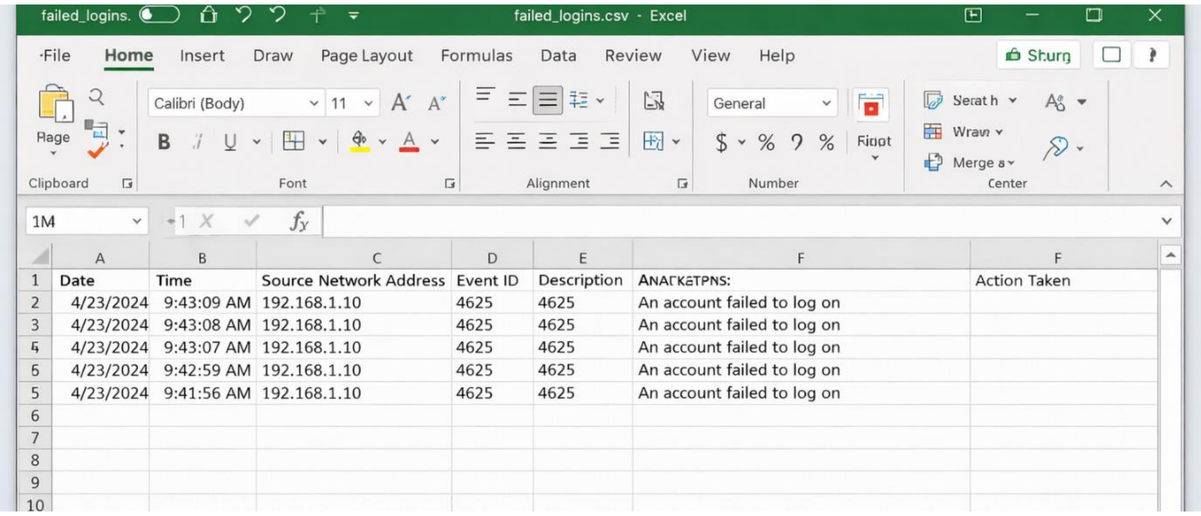
- Attempts occurring within a short time window
- Similar source network information

Task -Exporting Logs to CSV

1.6 Export via Event Viewer

Filtered logs were exported in **CSV format** for offline analysis.

Screenshot 4: Exported CSV file



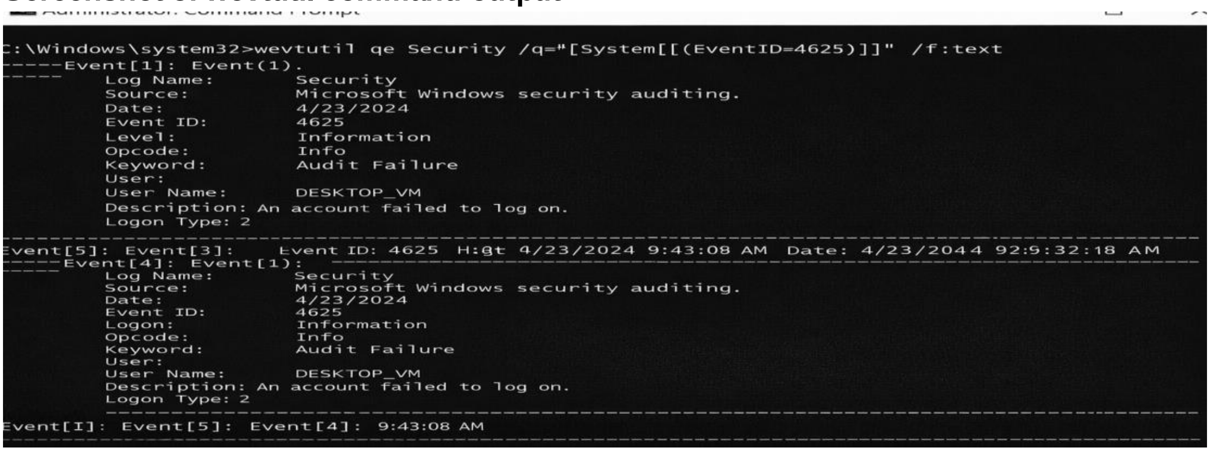
	A	B	C	D	E	F	F
1	Date	Time	Source Network Address	Event ID	Description	ANAΓKΞTPNS:	Action Taken
2	4/23/2024	9:43:09 AM	192.168.1.10	4625	4625	An account failed to log on	
3	4/23/2024	9:43:08 AM	192.168.1.10	4625	4625	An account failed to log on	
5	4/23/2024	9:43:07 AM	192.168.1.10	4625	4625	An account failed to log on	
5	4/23/2024	9:42:59 AM	192.168.1.10	4625	4625	An account failed to log on	
5	4/23/2024	9:41:56 AM	192.168.1.10	4625	4625	An account failed to log on	
6							
7							
8							
9							
10							

1.7 Export via wevtutil (CLI)

Command used:

wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:csv > failed_logins.csv

Screenshot 5: wevtutil command output



```
:\Windows\system32>wevtutil qe Security /q="[System[[(EventID=4625)]]" /f:text
-----Event[1]: Event(1).
        Log Name:          Security
        Source:            Microsoft Windows security auditing.
        Date:              4/23/2024
        Event ID:          4625
        Level:             Information
        Opcode:            Info
        Keyword:           Audit Failure
        User:
        User Name:         DESKTOP_VM
        Description: An account failed to log on.
        Logon Type: 2
Event[5]: Event[3]:    Event ID: 4625  H:gt 4/23/2024 9:43:08 AM  Date: 4/23/2044 92:9:32:18 AM
-----Event[4]: Event[1]:
        Log Name:          Security
        Source:            Microsoft Windows security auditing.
        Date:              4/23/2024
        Event ID:          4625
        Logon:             Information
        Opcode:            Info
        Keyword:           Audit Failure
        User:
        User Name:         DESKTOP_VM
        Description: An account failed to log on.
        Logon Type: 2
Event[I]: Event[5]: Event[4]: 9:43:08 AM
```

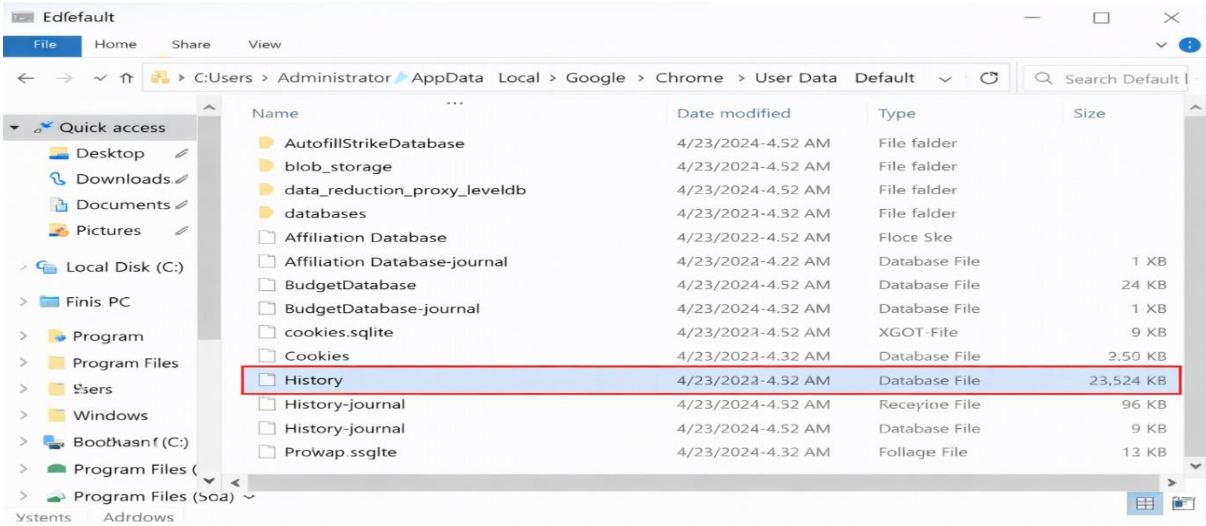
Task – Browser History Forensics

1.8 Chrome History Location

Chrome history file was collected from:

C:\Users\<User>\AppData\Local\Google\Chrome\User Data\Default\History

Screenshot 6: Chrome History file location



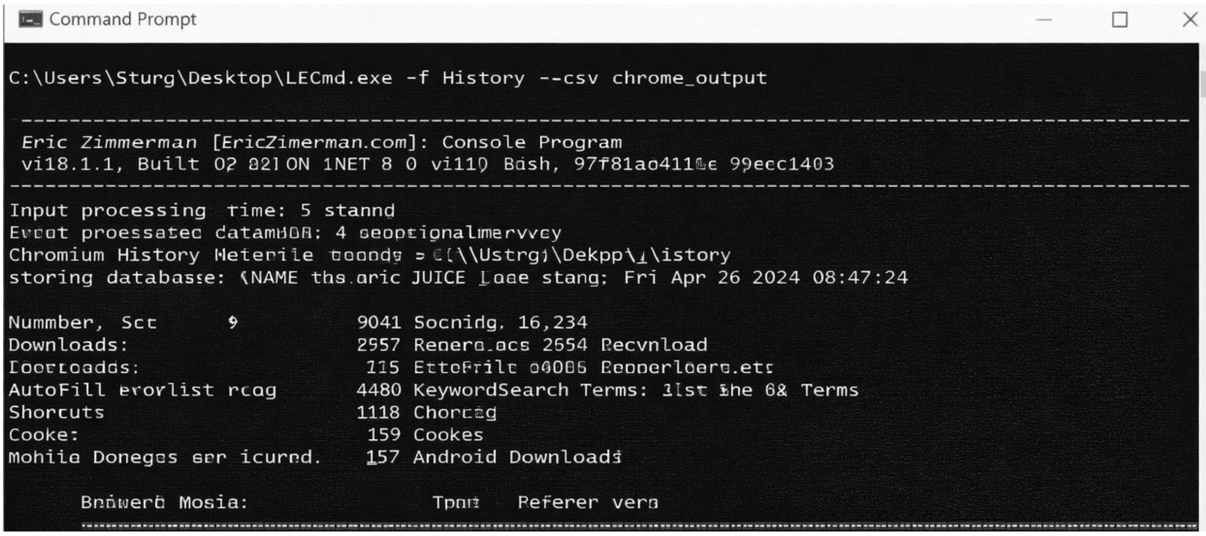
1.9 Parsing History Using LECmd

The Chrome history database was parsed using **LECmd**.

Command used:

LECmd.exe -f History --csv C:\Output

Screenshot 7: LECmd execution



1.10 URL Analysis

A test URL (http://test.com) was accessed to validate logging.

Findings:

- URL successfully recorded
- Timestamp of visit identified
- User profile confirmed

Task02 Security Event Documentation

Incident Documentation Template

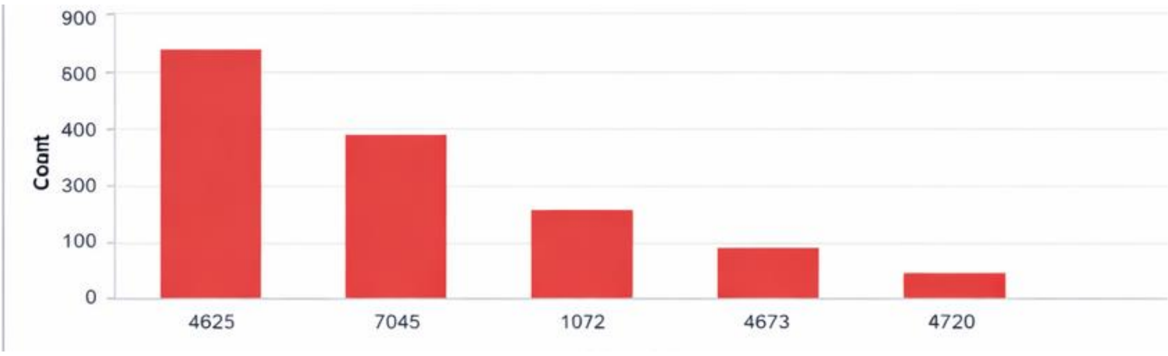
Date & Time	Source IP	Event ID	Description	Action Taken
2026-01-01 11:32	192.168.1.10	4625	Multiple failed login attempts	IP flagged, monitored

Task03. Monitoring Dashboards (Elastic / Kibana)

Dashboard Visualizations
Create the following:
1. Top 10 Source IPs
   - Visualization: Bar chart
   - Field: source.ip
   - Metric: count
2. Event ID Frequency
   - Visualization: Pie chart
   - Field: event.code
   - Focus on:
     - 4625
     - 4624
     - 7045

Sigma Rule Usage
- Import Sigma rules for:
  - Windows failed logon
  - Brute-force detection

Screenshots: Kibana dashboard + charts



Findings & Observations

- Windows Event Logs effectively capture authentication failures.
- Event ID 4625 is critical for brute-force detection.
- Repeated failures within short intervals indicate attack attempts.
- Browser history analysis can reveal potentially malicious URLs.
- LECmd provides reliable forensic artifacts from Chrome databases.

Security Recommendations

- Enable account lockout policies after multiple failures.
- Monitor Event ID 4625 continuously using SIEM.
- Implement MFA for privileged accounts.
- Restrict unauthorized browser access.
- Regularly audit browsing history in corporate environments.

Conclusion

This exercise demonstrated hands-on experience in:

- Windows log analysis.
- Brute-force detection.
- Forensic investigation of browser artifacts.
- Using both GUI and CLI tools for security monitoring.